

=== PQ-GNSS-TESLA End-to-End Baseband Simulator ===

[Part 1:Tx] Initialization & Signal Configuration...

[Part 2:Tx] CNAV-2 Data Generation & Dual TESLA Setup...

> Generating TESLA Chains (N=960, 24 hours duration)...

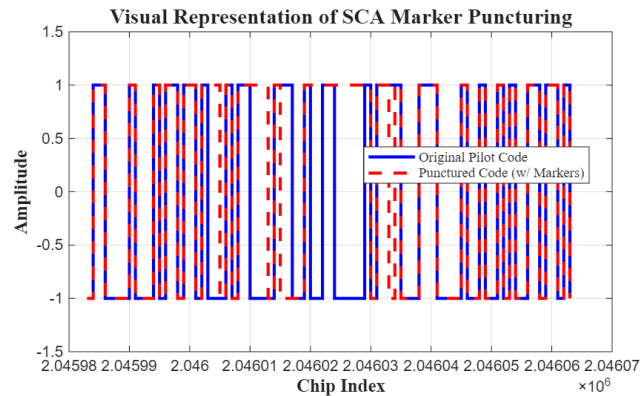
> Ground Segment: Generating Falcon-512 Keypair & Signature...

- Actual Signature size : 655 bytes

- Padded for OOB transmission: 666 bytes (5328 bits)

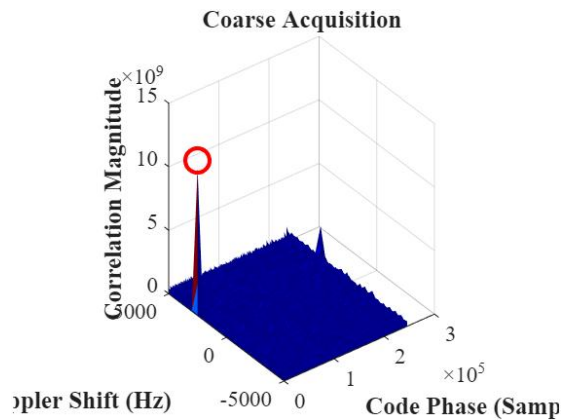
[Part 3:Tx] Frame 5 Overriding (Injecting K0, Z, MACs, Key)...

[Part 4:Tx] Inserting Dynamic SCA Markers into Pilot Channel...

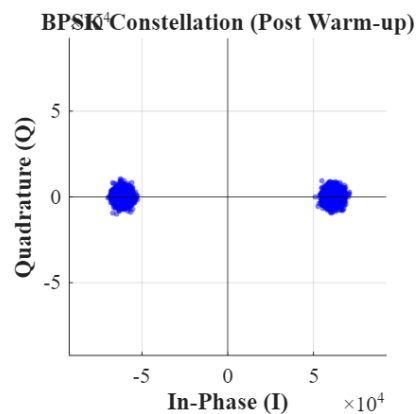


[Part 5:Rx] Coarse Signal Acquisition...

> Estimated Initial Doppler: 2500 Hz



[Part 6:Rx] Executing Phase-Locked Loop for 9210 Symbols...



[Part 7:Rx] CNAV-2 Decoding & TOI Recovery

- > Subframe 1 TOI (BCH) Recovered: 0

[Part 7.5:Rx] OOB Bootstrapping (Real Falcon-512 Verification)

- > Receiver: Downloading PQ Trust Anchor via OOB...
- > Receiver: Verifying Falcon-512 Signature...
- > [SUCCESS] Falcon-512 Signature Verified Mathematically!
- > TESLA Root Keys (K0) securely anchored. Receiver transitions to In-Band.

[Part 8:Rx] Data Authentication (Algorithm 1 & 2 Integration)

- > Executing Algorithm 2 (Continuous Verification & Rollover)...
- > Executing Algorithm 1 (Backward-Key-Chain Traversal)...
 - [SUCCESS] Algorithm 1: Disclosed key securely anchored to root!
- > 5-Frame Navigation MAC Verified (0/5 errors)
- > Data Channel Authentication Fully Successful!

[Part 9:Rx] Physical Layer Authentication (Chimera SCA)

- > Accumulated Soft-Decision Marker Energy: 54332006.85
- > SCA Authentication Passed! (Authentic Signal)